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Bandgap cell

Abstract

A self-starting bandgap cell for generating a reference voltage.

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Background/Summary

BACKGROUND

(1) The present disclosure relates to a bandgap cell for generating a reference voltage. In particular, the present disclosure relates to a bandgap cell that is self starting.

(2) FIG. 1 is a schematic of a known bandgap cell **100** with start-up circuitry **102**. The bandgap cell **100** comprises transistors **104**, **106**; a bipolar junction transistor (BJT) **108**; an amplifier **110** comprising BJT transistors **112**, **114** and additional circuitry **116**; and resistors **118**, **120**, **122**. A bandgap cell, such as the bandgap cell **100**, may be used to generate a reference voltage V_{bg} on a chip.

(3) If no current is flowing in the bandgap cell **100** then the system will do nothing, and no reference voltage V_{bg} will be generated. A start-up process is used, as provided by the start-up circuitry **102**, where a bias current I_{bias} is injected into the bandgap cell **100**. Current will build up in the system and reach a stable settling point. At this point the cell will operate and a reference voltage will be generated.

(4) The start-up circuitry **102** may not function well when the bandgap cell **100** is operable using nanoampere currents (for example when the operating current is less than 100 nA. This is because the start-up circuitry **102** cannot easily detect nanoampere current levels in the main bandgap cell **100** meaning that the start-up circuit **102** can get activated when the bandgap cell **100** is already in operation, and the circuitry **102** may remain engaged, thereby interfering with main circuit operation, or it may erroneously detect small current in the bandgap cell and not engage at all and the bandgap cell will fail to start-up

SUMMARY

(5) It is desirable to provide a bandgap cell for generating a reference voltage that reduces the issues of known start-up techniques.

(6) According to a first aspect of the disclosure there is provided a self-starting bandgap cell for generating a reference voltage.

- (7) Optionally, the self-starting bandgap cell is configured to provide the reference voltage.
- (8) Optionally, the self-starting bandgap cell comprises a negative threshold transistor configured to provide an operating current when coupled to a supply voltage, the reference voltage being generated when the operating current flows in the self-starting bandgap cell.
- (9) Optionally, the negative threshold transistor is an NMOS transistor or a PMOS transistor.
- (10) Optionally, the self-starting bandgap cell comprises an amplifier comprising a first input terminal, a second input terminal and an output terminal, the output terminal being coupled to a gate of the negative threshold transistor, and a resistor string comprising a plurality of resistive elements coupled in series with the negative threshold transistor, the resistor string being coupled to the first and second input terminals of the amplifier.
- (11) Optionally, the plurality of resistive elements comprises a first resistor, a second resistor and a third resistor.
- (12) Optionally, one or both of the second and third resistors comprises a variable resistor.
- (13) Optionally, the second resistor is coupled to the first resistor at the first input terminal of the amplifier and the first resistor is coupled to the third resistor at the second input terminal of the amplifier.
- (14) Optionally, the self-starting bandgap cell comprises a first circuit component coupled to the resistor string, the first circuit component comprising a diode, an NPN bipolar junction transistor or a PNP bipolar junction transistor.
- (15) Optionally, the amplifier comprises a BJT input pair comprising a first input bipolar junction transistor, and a second input bipolar junction transistor, wherein the first input bipolar junction transistor is coupled to one of the first input terminal and the second input terminal and the second input bipolar junction transistor is coupled to the other of the first input terminal and the second input terminal, and the BJT input pair is configured to receive an amplifier bias current.
- (16) Optionally, the BJT input pair is configured to provide a first voltage at the first input terminal of the amplifier and a second voltage at the second input terminal of the amplifier.
- (17) Optionally, the BJT input pair is configured to have a BJT area ratio to provide the first voltage at the first input terminal of the amplifier and the second voltage at the second input terminal of the amplifier.
- (18) Optionally, the BJT area ratio is 1:8 or 1:4 or 2:8 or 1:16 or 2:16.
- (19) Optionally, the self-starting bandgap cell is configured to receive the amplifier bias current from external circuitry.
- (20) Optionally, the first component comprises the diode or the PNP bipolar junction transistor, and the first and second input bipolar junction transistors are PNP bipolar junction transistors.
- (21) Optionally, the first component comprises a first terminal coupled to the third resistor and a second terminal coupled to the negative threshold transistor at an output node, the reference voltage being provided at the output node during operation, a base of the first input bipolar junction transistor is coupled to the first input terminal, and a base of the second input bipolar junction transistor is coupled to the second input terminal.
- (22) Optionally, the BJT input pair is configured to provide a first voltage at the first input terminal of the amplifier and a second voltage at the second input terminal of the amplifier. Optionally, the BJT input pair is configured to have an offset ratio to provide the first voltage at the first input terminal of the amplifier and the second voltage at the second input terminal of the amplifier. Optionally, the BJT area ratio is 1:8 or 1:4 or 2:8 or 1:16 or 2:16.
- (23) Optionally, the first component comprises the diode or the NPN bipolar junction transistor, and the first and second input bipolar junction transistors are NPN bipolar junction transistors.
- (24) Optionally, the first component comprises a first terminal coupled to the second resistor and a second terminal coupled to a voltage terminal, the third resistor is coupled to the negative threshold transistor at an output node, the reference voltage being provided at the output node during operation, a base of the first input bipolar junction transistor is coupled to the second input

terminal, and a base of the second input bipolar junction transistor is coupled to the first input terminal.

(25) Optionally, the BJT input pair is configured to provide a first voltage at the first input terminal of the amplifier and a second voltage at the second input terminal of the amplifier. Optionally, the BJT input pair is configured to have a BJT area ratio to provide the first voltage at the first input terminal of the amplifier and the second voltage at the second input terminal of the amplifier.

(26) Optionally, the offset ratio is 1:8 or 1:4 or 2:8 or 1:16 or 2:16.

(27) According to a second aspect of the disclosure there is provided a method of generating a reference voltage using a self-starting bandgap cell.

(28) It will be appreciated that the method of the second aspect may include using and/or providing any of the features of the first aspect, or any other features as described herein, and in accordance with the understanding of the skilled person.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is described in further detail below by way of example and with reference to the accompanying drawings in which:

(2) FIG. 1 is a schematic of a known bandgap cell with start-up circuitry;

(3) FIG. 2(a) is a schematic of a self-starting bandgap cell in accordance with a first embodiment of the present disclosure, FIG. 2(b) is a schematic of a self-starting bandgap cell in accordance with a second embodiment of the present disclosure;

(4) FIG. 3(a) is a schematic of a self-starting bandgap cell in accordance with a third embodiment of the present disclosure, FIG. 3(b) is a schematic of a bandgap cell in accordance with a fourth embodiment of the present disclosure;

(5) FIG. 4(a) is a schematic of a specific embodiment of the amplifier in accordance with a fifth embodiment of the present disclosure, FIG. 4(b) is a schematic of a specific embodiment of the amplifier in accordance with a sixth embodiment of the present disclosure, FIG. 4(c) is a schematic of a bias current generator; and

(6) FIG. 5(a) is a schematic of a self-starting bandgap cell in accordance with a seventh embodiment of the present disclosure, FIG. 5(b) is a schematic of a self-starting bandgap cell in accordance with an eighth embodiment of the present disclosure, FIG. 5(c) is a schematic of a self-starting bandgap cell in accordance with a ninth embodiment of the present disclosure.

DETAILED DESCRIPTION

(7) A band gap cell is used to generate a reference voltage that is proportional to the bandgap energy of silicon. This reference voltage is independent of supply and can have a very low-temperature coefficient. The standard bandgap cell has two stable operating points. In the operating point, a non-zero bias current level exists where bandgap voltage is available. The other stable operating point is with zero current and the cell remains off. This is why, in known systems, a startup circuit is added to inject the initial bias current and allow the circuit to come out of its zero bias current state and settle into desired bias current state. Once it reaches there, the bandgap cell can maintain its own bias current.

(8) FIG. 2(a) is a schematic of a self-starting bandgap cell **200** for generating a reference voltage V_{bg} in accordance with a first embodiment of the present disclosure. The self-starting bandgap cell **200** may be configured to provide the reference voltage V_{bg} when the cell is enabled.

(9) The bandgap cell **200** is self-starting and therefore eliminates the requirement for auxiliary self-starting circuitry, such as the circuitry **102** as shown in FIG. 1. This means that there is reduced power consumption and reduced chip area when compared with the system **100**, as the need for additional circuitry is removed. Also, the system is highly reliable. It will neither fail to start-up nor

generate a reference voltage that is not accurate due to interference from the start-up circuit.

(10) FIG. 2(b) is a schematic of a self-starting bandgap cell **201** comprising a negative threshold transistor **202**, in accordance with a second embodiment of the present disclosure. The negative threshold transistor **202** is configured to provide the operating current I_{op} when coupled to a supply voltage v_{dd} . The supply voltage v_{dd} may, for example, be within the range 1.5V to 5.5V. The reference voltage V_{bg} is generated when the operating current I_{op} flows in the self-starting bandgap cell **201**, during operation. The negative threshold transistor **202** may be an NMOS transistor. It is possible to use a negative or zero threshold PMOS in this position trying to inject current in the loop, in that polarity of feedback will need changes because a PMOS will add inversion in the loop. A negative threshold transistor is a transistor that will conduct a current even when a voltage of 0V is applied across its gate and source.

(11) FIG. 3(a) is a schematic of a self-starting bandgap cell **300** in accordance with a third embodiment of the present disclosure. The bandgap cell **300** comprises an amplifier **302** comprising input terminals **304**, **306** and an output terminal **308**. The output terminal **308** is coupled to a gate **310** of the negative threshold transistor **202**. The bandgap cell **300** further comprises a resistor string **312** comprising a plurality of resistive elements coupled in series with respect to each other and the negative threshold transistor **202**. The resistor string **312** is coupled to the input terminals **304**, **306**.

(12) The bandgap cell **300** may further comprises a circuit component **314** that is coupled to the resistor string **312**. The circuit component **314** may comprise a diode, an NPN bipolar junction transistor or a PNP bipolar junction transistor.

(13) FIG. 3(b) is a schematic of a bandgap cell **316** in accordance with a fourth embodiment of the present disclosure. In the present embodiment the plurality of resistive elements comprises resistors **318**, **320**, **322**. One or both of the resistors **318**, **322** may be variable resistors to enable trimming functionality to be provided. In the present embodiment, both the resistors **318**, **322** comprise variable resistors. The resistor **318** is coupled to the resistor **320** at the input terminal **304** and the resistor **320** is coupled to the resistor **322** at the input terminal **306**.

(14) FIG. 4(a) is a schematic of a specific embodiment of the amplifier **302** in accordance with a fifth embodiment of the present disclosure. The amplifier **302** comprises a BJT input pair **400** comprising input bipolar junction transistors **402**, **404**. The transistor **402** is coupled to the input terminal **304** and the transistor **404** is coupled to the input terminal **306**. The BJT input pair **400** is configured to receive an amplifier bias current I_{bias} . A bias current typically refers to a current that is generated and provided to other circuitry, for example on a chip, that requires the bias current for its operation. In the present example, the bias current I_{bias} is required for the operation of the amplifier **302**. In this topology BJT's will be of PNP type.

(15) The BJT input pair **400** may be configured to provide a voltage V_1 at the input terminal **304** and a voltage V_2 at the input terminal **306**. This functionality may be provided by a BJT area ratio applied to the transistors **402**, **404**. The ratio may be 1:8 for the transistor **402**: the transistor **404**. It will be appreciated that in further embodiments other ratios may be used such as 1:4, 2:8, 1:16 or 2:16.

(16) If both BJT **402**, **404** are of the same size, the error amplifier loop will force V_1 and V_2 to be equal. But if BJT **404** is bigger in size than BJT **402** (8 times bigger in the present example), then V_2 will be forced higher than V_1 by the error amplifier loop. Then $(V_2 - V_1)$ will be a finite voltage which may be called an offset voltage.

(17) FIG. 4(b) is a schematic of a specific embodiment of the amplifier **302** in accordance with a sixth embodiment of the present disclosure. In the present embodiment, the transistor **402** is coupled to the input terminal **306** and the transistor **404** is coupled to the input terminal **304**. In this topology BJT's will be of NPN type. In FIG. 4(b), transistor **404** is bigger in size than **402**.

(18) FIG. 4(c) is a schematic of a bias current generator **406** that may function as external circuitry

for generating and then providing the bias current I_{bias} to the amplifier **302**. The bias current generator **406** comprises a negative threshold transistor **408** and a poly resistor **410**.

(19) The bias current generator **406** is an example circuit for providing the amplifier bias current I_{bias} . The bias current generator **406** can generate a supply independent current. It uses a negative threshold NMOS device (reference numeral **408**) with a resistor (reference numeral **410**). Current flowing in the negative threshold transistor **408** can, for example, be mirrored and supplied to the amplifier **302** as the bias current I_{bias} .

(20) As illustrated in FIG. **1**, existing systems require start-up circuitry (for example as provided by the circuit **102**) for injecting the initial bias current and once start up is complete bandgap cell will generate its own bias current at a stable operating point I_{bias} for the amplifier **110**. In the present example, start-up circuitry is not required. It will be appreciated that there are other means of bias current generation that may be used, also without requiring start-up circuitry, and in accordance with the understanding of the skilled person.

(21) FIG. **5(a)** is a schematic of a self-starting bandgap cell **500** in accordance with a seventh embodiment of the present disclosure. The component **314** may comprises a diode or a PNP bipolar junction transistor, with the input bipolar junction transistors being PNP bipolar junction transistors. In the present embodiment, the first component **314** comprises the PNP bipolar junction transistor.

(22) The component **314** comprises a terminal **502** coupled to the resistor **322** and a terminal **504** coupled to the negative threshold transistor **202** at an output node N_{out} . The reference voltage V_{bg} is provided at the output node N_{out} during operation of the bandgap cell **500**. A base of the transistor **402** is coupled to the input terminal **304** and a base of the transistor **404** is coupled to the input terminal **306**.

(23) The amplifier **302** further comprises additional circuitry **501** being coupled to the BJT input pair **402**, **404** and the output terminal **308** of the amplifier **302**. The additional circuitry **501** may be implemented as a folded cascode amplifier which is well-known in the art. In further embodiments, other differential amplifier topologies may be used in accordance with the understanding of the skilled person.

(24) FIG. **5(b)** is a schematic of a self-starting bandgap cell **506** in accordance with an eighth embodiment of the present disclosure. The component **314** may comprises a diode or a NPN bipolar junction transistor, with the input bipolar junction transistors being NPN bipolar junction transistors. In the present embodiment, the first component **314** comprises the NPN bipolar junction transistor.

(25) The component **314** comprises a terminal **508** coupled to the resistor **318** and a terminal **510** coupled to a voltage terminal **512** at a voltage v_{ss} . The resistor **322** is coupled to the negative threshold transistor **202** at the output node N_{out} . A base of the transistor **404** is coupled to the input terminal **304** and a base of the transistor **402** is coupled to the input terminal **306**.

(26) The embodiment presented in FIG. **5(a)** use a BJT input pair **402**, **404** with a BJT area ratio of 1:8 to generate the voltages V_1 , V_2 to provide ΔV_{BE} (labelled as “Delta V_{BE} ” in the Figure) which is the voltage across the resistor **320**. V_{BE} , as shown on the Figure, denotes the base to emitter voltage of the BJT **314**, which comprises a PNP BJT in FIG. **5(a)**, as discussed previously. A supply independent bias current I_{bias} is provided to the amplifier **302** from external circuitry (such as the bias current generator **406**). It will be appreciated that in further embodiments other ratios may be used such as 1:4, 2:8, 1:16 or 2:16.

(27) The use of the supply independent bias current I_{bias} from external circuitry, the negative threshold transistor **202** coupled to the supply voltage v_{dd} , and the addition of the PNP BJT **314** positioned as shown in the Figure contribute to the provision of the operating current I_{op} which enables self-starting of the bandgap cell **500**. The resistors **318**, **320**, **322** are used to generate the reference voltage V_{bg} (which may be referred to as the “bandgap voltage”), and may be represented as follows:

$$(28) \quad V_{bg} = ((V_{BE} / R_1) \times (R_1 + R_2 + R_3)) + V_{BE} \quad (1)$$

where R_1 , R_2 and R_3 denote the resistances of resistors **320**, **318** and **322**, respectively.

(29) In a specific embodiment, R_2 and R_3 are of equal value and are trimmable (for example, as provided by the variable resistors). This arrangement allows cancellation of errors due to base current flowing into the amplifier BJTs **402**, **404**.

(30) If this is not done bandgap voltage will have offset voltage. And this offset voltage will vary with temperature.

(31) Since the negative threshold NMOS N_1 (labelled **202**) is a top device (positioned in an upper portion of the circuit and coupled to the supply voltage v_{dd}), it will inject current in the output branch (formed by transistors **202**, **314** and resistors **318**, **320**, **322**) whenever a valid v_{dd} supply is present.

(32) We have introduced a PNP BJT B_1 (labelled **314**) in the output branch before we go to amplifier **302** inputs V_1 and V_2 . This means that amplifier **302** inputs V_1 and V_2 will see a potential less than equal to $(V_{dd} - V_{BE})$. $(V_{dd} - V_{BE})$ max voltage means amplifier input pair B_2 and B_3 always turns on.

(33) Since the input pair B_2 (labelled **404**) and B_3 (labelled **402**) are PNP BJT, this means they will always turn on. Because even if emitter of B_2 is at V_{dd} , the base of B_2 will be $(V_{dd} - V_{BE})$. This means even if we ignore voltage drop in R_1 , R_2 , R_3 $V_{be2} = (V_{dd} - V_{BE})$ and similarly, $V_{be3} = (V_{dd} - V_{BE})$. This is sufficient to guarantee input pairs B_2 and B_3 will always turn on. Since N_1 (labelled **202**) is guaranteed to inject some current there will be an IR drop in R_3 R_1 and R_2 , this further helps in turn on of B_2 and B_3 because V_{be2} and V_{be3} will further increase as IR drop increases. V_{be2} and V_{be3} denotes base to emitter voltage of the transistor **404** and **402**, respectively. Thus, the bandgap cell **500** does not require a startup circuit, it is self-starting, and it will always work once the supply V_{dd} is applied.

(34) Since the bias current I_{bias} to the amplifier **302** is supplied externally from another circuit, this further eliminates the need for a start-up circuit as we are not self-biasing the amplifier as is used in known systems. Self-biasing is a technique where i_{bias_main} current flowing in one of the branches of the amplifier loop is mirrored back into its tail current i_{bias} . This has a drawback that if i_{bias_main} current is negligible its mirrored version can also be negligible and amplifier will not work causing startup failure. Therefore, embodiments of the present disclosure using external circuitry for the amplifier bias current I_{bias} resolves this issue in known systems.

(35) Further embodiments may alter the arrangements of the resistors **318** and **322** without affecting the self-starting operation. However, changing arrangement of resistors **318** and **322** may cause an offset voltage in the generated bandgap voltage V_{bg} due to base currents of BJT's.

(36) FIG. 5(b) demonstrates similar behaviour to as described in FIG. 5(a), as will be clear to the skilled person. For the bandgap cell **506**, and compared with the circuit **500**, there is provided self-starting behaviour with NPN BJT input pair B_2 and B_3 (labelled **402**, **404**). Here, B_1 (labelled **314**) has been moved to the bottom of the ladder.

(37) Here N_1 (labelled **202**) makes sure a current is always injected in B_1 (labelled **314**) and thus producing V_{BE} which is sufficient to turn ON B_2 and B_3 (labelled **404** and **402**, respectively). Voltage drop across resistor **318** and resistor **320** further contributes to strong turn ON of B_2 and B_3 (labelled **404** and **402**, respectively).

(38) FIG. 5(c) is a schematic of a self-starting bandgap cell **514** having a specific embodiment of the error amplifier **302** comprising a folded cascode topology, in accordance with a ninth embodiment of the present disclosure.

(39) In summary, self starting functionality is enabled by: The negative threshold device **202** at the top The correct proposed location of the component **314** The externally generated bias current I_{bias} of the amplifier **302**

(40) In summary, in known systems, auxiliary startup circuits are required to startup a bandgap cell.

While it is okay to use auxiliary startup circuits when the bias current of the bandgap cell is in μA range it becomes risky and difficult for a nano ampere bandgap cell. Specifically, in nA applications traditional start up cells cannot detect nA current levels flowing in main amplifier loop meaning that they can get falsely activated and remain engaged thereby interfering with main circuit operation. Furthermore, auxiliary startup circuitry occupies chip area.

(41) Embodiments of the present disclosure provide the following advantages; There is provided a robust architecture that does not require a startup circuit, thereby reducing occupied chip area as would otherwise be required for auxiliary startup circuitry. Proposed circuit is extremely useful for ultra-low power circuits as it does not require any additional power and it is guaranteed to start. It works for both nA and μA level circuits so it is a versatile reusable architecture. Startup circuits provide an area overhead and might use some current all the time during operation. For example for a 5 nA bandgap cell even 1 nA in startup will waste 20% of overall power budget. Proposed circuit is extremely useful for ultra-low power circuits as it does not take any additional power as would otherwise be wasted in startup cells. It has excellent power supply rejection because the additional circuitry compared with known systems does not compromise power supply rejection property of the bandgap cell.

(42) Common reference numerals or variables between Figures represent common features.

(43) Various improvements and modifications may be made to the above without departing from the scope of the disclosure.

Claims

1. A self-starting bandgap cell for generating a reference voltage, the self-starting bandgap cell comprises: a negative threshold transistor configured to provide an operating current when coupled to a supply voltage, the reference voltage being generated when the operating current flows in the self-starting bandgap cell; an amplifier comprising a first input terminal, a second input terminal and an output terminal, the output terminal being coupled to a gate of the negative threshold transistor, wherein the amplifier comprises a BJT input pair comprising: a first input bipolar junction transistor; and a second input bipolar junction transistor; a resistor string comprising a plurality of resistive elements coupled in series with the negative threshold transistor, the resistor string being coupled to the first and second input terminals of the amplifier, wherein the plurality of resistive elements comprises a first resistor, a second resistor and a third resistor; and a first circuit component coupled to the resistor string, wherein: the first input bipolar junction transistor is coupled to one of the first input terminal and the second input terminal and the second input bipolar junction transistor is coupled to the other of the first input terminal and the second input terminal; the BJT input pair is configured to receive an amplifier bias current; the first circuit component comprises a first terminal coupled to the third resistor and a second terminal coupled to the negative threshold transistor at an output node, the reference voltage being provided at the output node during operation; a base of the first input bipolar junction transistor is coupled to the first input terminal; and a base of the second input bipolar junction transistor is coupled to the second input terminal.
2. The self-starting bandgap cell of claim 1, wherein one or both of the second and third resistors comprises a variable resistor.
3. The self-starting bandgap cell of claim 1, wherein the second resistor is coupled to the first resistor at the first input terminal of the amplifier and the first resistor is coupled to the third resistor at the second input terminal of the amplifier.
4. The self-starting bandgap cell of claim 3, comprising a first circuit component comprising a diode, an NPN bipolar junction transistor or a PNP bipolar junction transistor.
5. The self-starting bandgap cell of claim 4, wherein: the first circuit component comprises the diode or the PNP bipolar junction transistor; and the first and second input bipolar junction

transistors are PNP bipolar junction transistors.

6. The self-starting bandgap cell of claim 4, wherein: the first circuit component comprises the diode or the NPN bipolar junction transistor; and the first and second input bipolar junction transistors are NPN bipolar junction transistors.

7. The self-starting bandgap cell of claim 6, wherein: the first circuit component comprises a first terminal coupled to the second resistor and a second terminal coupled to a voltage terminal; the third resistor is coupled to the negative threshold transistor at an output node, the reference voltage being provided at the output node during operation; a base of the first input bipolar junction transistor is coupled to the second input terminal; and a base of the second input bipolar junction transistor is coupled to the first input terminal.

8. The self-starting bandgap cell of claim 1, wherein the BJT input pair is configured to provide a first voltage at the first input terminal of the amplifier and a second voltage at the second input terminal of the amplifier.

9. The self-starting bandgap cell of claim 8, wherein the BJT input pair is configured to have a BJT area ratio to provide the first voltage at the first input terminal of the amplifier and the second voltage at the second input terminal of the amplifier.

10. The self-starting bandgap cell of claim 9, wherein the BJT area ratio is 1:8 or 1:4 or 2:8 or 1:16 or 2:16.

11. The self-starting bandgap cell of claim 9, wherein the BJT input pair is configured to have an offset ratio to provide the first voltage at the first input terminal of the amplifier and the second voltage at the second input terminal of the amplifier.

12. The self-starting bandgap cell of claim 11, wherein the BJT area ratio is 1:8 or 1:4 or 2:8 or 1:16 or 2:16.

13. The self-starting bandgap cell of claim 1, configured to receive the amplifier bias current from external circuitry.

14. A method comprising generating the reference voltage using the self-starting bandgap cell of claim 1.
